

2024, Mar. 2024 Release LOC:Acute Adult Hospital in the Home

Overview

Select Day

Episode Day 1

Episode Day 2-X ⁽⁷⁰⁾

Notes

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Overview

+Reviewer Instructional Note

Instruction: This subset is for the admission review of patients with the following conditions who are participating in a Hospital in the Home (HITH) program:

- Cellulitis
- Chronic obstructive lung disease (COPD)
- Heart failure (HF)
- Pneumonia
- Pyelonephritis

For continued stay review, refer to the appropriate episode day at the Acute level of care within the corresponding condition-specific subset.

Hospital in the Home (HITH) is an alternative Acute level of care that brings community-based services usually associated with inpatient hospitalization into a patient's home. Patients are triaged, diagnosed, and admitted by their primary care physician or an emergency department (ED) physician or, in another model of HITH, they may be discharged early from a hospital to continue HITH care at home (Federman et al., JAMA Internal Medicine 2018;; Levine et al., J Gen Intern Med 2018, 33: 729-36; Shepperd et al., Cochrane Database Syst Rev 2016, 9: Cd007491; Cryer et al., Health Aff (Millwood) 2012, 31: 1237-43). HITH has been found to be an effective alternative to inpatient care, with increased patient satisfaction rates, comparable or reduced mortality and readmission rates, and lower costs for selected patient populations (Levine et al., J Gen Intern Med 2018, 33: 729-36; Conley et al., JAMA Intern Med 2016, 176: 1693-702; Caplan, Med J Aust 2013, 198: 195-6; Cryer et al., Health Aff (Millwood) 2012, 31: 1237-43).

HITH provides 24h physician and registered nurse oversight. Other services (physical therapy, occupational therapy, meal delivery, home health aides) are available based on patient needs. Patient-provider communications are managed through secure phone and video technology. Services provided may include oxygen, IV fluids, antibiotics, respiratory therapy including nebulizers, IV pumps, X-rays, and point of care diagnostics. While in the home, patients may receive temperature, heart rate/telemetry, respiratory rate, oximetry, mobility, or sleep monitoring. Patients participating in a HITH program may also be transported to the hospital briefly for services that are unable to be provided in the home setting, such as computerized tomography (CT), magnetic resonance imaging (MRI), and/or endoscopic procedures such as esophagogastroduodenoscopy (EGD).

Patient eligibility is based on primary diagnosis and program inclusions/exclusion criteria including:

- Willingness to participate in the program
- A safe home environment within the geographical catchment area of the HITH program
- Adequate utilities (heat and/or air conditioning, if necessary, and running water)
- Established IV access
- Ability to ambulate and transfer independently to a bedside commode or with the assistance of a caregiver

Some HITH programs transition their patients to home care as their acute care needs decrease. In a 2018 observational

study that bundled HITH care with 30-day episodes of postacute home-based transitional care, patients were found less likely to be readmitted to hospitals, revisit EDs, or transfer to skilled nursing facilities (SNFs). They also experienced a shorter acute care period or length of stay and rated communications with providers higher than inpatient hospitalized patients (Federman et al., JAMA Internal Medicine 2018:). Patients who participate in HITH programs whose conditions are deteriorating may require transfer to an acute care hospital.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Episode Day 1, One:**

(Symptom or finding within 24h)

● **HOSPITAL IN THE HOME, Both:**● Patient selection, **All:** ^(1, 2)

- Patient agrees to participate in Hospital in the Home program
- Home environment conducive to safe care ⁽³⁾

● Clinically stable for home management, **All:**● O₂ sat, **One:** ⁽⁴⁾

- ≥ 90%(0.90) or baseline and on supplemental oxygen < 40%(0.40) ⁽⁵⁾
- ≥ 90%(0.90) or baseline and supplemental oxygen not required

- Systolic blood pressure ≥ 90 mmHg

- Mental illness absent or controlled ^(6, 7)- Substance use disorder absent or controlled ⁽³⁾● Finding, **Both:** ^(8, 9)

- Hospital in the Home (HITH) screening complete and patient accepted

● Condition, **One:** ^(2, 8)● Cellulitis, **Both:** ⁽¹⁰⁾● Finding, ≥ **One:**

- Immunocompromised ⁽¹¹⁾

- Located over a prosthesis or implanted device ⁽¹²⁾

- Anti-infective ⁽¹³⁾

● COPD, **All:** ^(14, 15, 16, 17, 18)

- Short-acting beta-agonist administered prior to admission by licensed medical professional ≥ 2 doses ⁽¹⁹⁾

● Post treatment finding, ≥ **One:**● Increased work of breathing, ≥ **One:** ⁽²⁰⁾

- Unable to take PO due to respiratory effort
- Hunched over position

● High risk for adverse event, **Both:**● Post treatment finding, ≥ **One:**

- O₂ sat 90-91%(0.90-0.91) and < baseline ⁽²¹⁾

● Increased work of breathing, ≥ **One:**

- Difficulty taking PO due to respiratory effort
- Prefers sitting due to respiratory effort ⁽²²⁾
- Talks in phrases due to respiratory effort

● Clinical risk factor, ≥ **One:**● Finding, ≥ **One:** ⁽²³⁾

- Blood urea nitrogen (BUN) > 25.0 mg/dL(8.9 mmol/L)

- Mental status changes (excludes coma, stupor, or obtundation) ⁽²⁴⁾

- Home oxygen therapy ≥ 15 h/24h ⁽²⁵⁾

- Cor pulmonale ^(26, 27)

- Heart failure, NYHA Class III or Class IV ^(28, 29, 30)

● Pneumonia actual or suspected and, ≥ **One:** ⁽³¹⁾

- Temperature ≥ 100.4°F(38.0°C) ⁽³²⁾

- WBC ≥ 12,000/cu.mm(12x10⁹/L)

- WBC ≤ 4,000/cu.mm(4 x10⁹/L)

- Bands > 10%(0.10)

● Intervention, **Both:** ⁽³³⁾

- Bronchodilator ≥ 4x/24h (includes inhaled) ^(34, 35)

- Corticosteroid (includes PO or inhaled) ⁽³⁶⁾

● Heart failure (excludes new onset), **Both:** ^(30, 37)

● **Dyspnea, not returned to baseline after at least 1 dose of diuretic and ≥ 2 h treatment and, $\geq$ One:** ^(21, 38, 39)

● **Inadequate diuresis, $\geq$ One:** ⁽⁴⁰⁾

- Urine sodium < 50 mmol/L at 2h after diuretic ⁽⁴¹⁾
- Urine output < 150 ml/h at 2h after diuretic ⁽⁴¹⁾

● **Clinical risk factor, $\geq$ One:** ⁽⁴²⁾

- Creatinine ≥ 1.5 x baseline or estimated baseline ^(43, 44)
- Chronic kidney disease (excludes chronic dialysis) and creatinine ≥ 2.75 mg/dL ⁽⁴⁵⁾
- BUN ≥ 43 mg/dL ⁽⁴⁶⁾
- Sodium < 130 mEq/L(130 mmol/L) ⁽⁴⁷⁾

● **Diuretic, $\geq$ One:** ^(48, 49)

- Diuretic ≥ 2 doses
- Escalation or change of diuretic

● **Pneumonia, All:** ⁽⁵⁰⁾

- Pneumonia confirmed by imaging ^(31, 51)

● **Finding, $\geq$ One:**

- ≥ 2 lobes ⁽⁵²⁾

● **High risk for adverse event, Both:**

- O₂ sat 90-91%(0.90-0.91) and $<$ baseline ⁽²¹⁾

● **Clinical risk factor, $\geq$ One:** ⁽⁵³⁾

- COPD, Grade 3 or 4 (current) ⁽⁵⁴⁾
- Heart failure, NYHA Class III or IV (current) ^(28, 30, 55)

● **Intervention, $\geq$ One:** ⁽⁵⁶⁾

- Anti-infective ⁽⁵⁷⁾

● **PO anti-infective and, $\geq$ One:** ⁽⁵⁷⁾

- IV fluid ⁽⁵⁸⁾
- Requiring supplemental oxygen ⁽⁵⁾

● **Pyelonephritis or Complex UTI, All:** ⁽⁵⁹⁾

● **Urinary symptoms with abnormal urinalysis and, $\geq$ One:** ^(60, 61)

- ≥ 10 WBC/mm³ ⁽⁶²⁾
- Leukocyte esterase positive ⁽⁶³⁾
- Nitrite positive ⁽⁶⁴⁾

● **Risk factor, $\geq$ One:**

- Pregnancy and ≥ 24 wks gestation ⁽⁶⁵⁾
- Urinary stent by history ⁽⁶⁶⁾
- Urinary tract obstruction or anatomical abnormality by imaging or history ⁽⁶⁷⁾

● **Urine culture pending and, Both:**

- Anti-infective ^(68, 69)
- IV fluid ⁽⁵⁸⁾

● **Episode Day 2-X, One:** ⁽⁷⁰⁾

● **HOSPITAL IN THE HOME, One:**

● **Not clinically stable for discharge and requires continued stay, One:**

- Cellulitis (**use Infection: Cellulitis**)
- COPD (**use COPD subset**)
- Heart failure (**use Heart Failure subset**)
- Pneumonia (**use Infection: Pneumonia subset**)
- Pyelonephritis or Complex UTI (**use Infection: Pyelonephritis or Complex UTI subset**)

Notes:**1:**

Hospital in the Home (HITH) or Hospital at Home is an Acute level of care model that has been researched and tested in the United States since the late 1990s. In a landmark 1997 publication, Leff et al., developed patient and condition-specific selection criteria for a Hospital at Home care model with nursing supervision, physician visits, ancillary therapies, point of care diagnostics, and monitoring technologies (Leff et al., J Am Geriatr Soc 1997, 45: 1066-73).

2:

Hospital in the Home (HITH) programs carefully screen candidates, considering inherent capabilities of their programs as well as each patient's goals of care.

3:

A safe home environment includes adequate utilities (e.g., running water, heating and/or air conditioning as needed), accessibility to care (e.g., within program's geographical catchment area and Home Care services), and a safe environment for providers to administer services. Patients may be excluded from HITH programs if they live alone and do not have a reliable caregiver, or if there is a history of domestic abuse, active substance use disorder or daily methadone pickup requirement, or they are in police custody (Centers for Disease Control and Prevention, Occupational Hazards in Home Healthcare. 2018).

4:

InterQual® consultants agree that Hospital in the Home (HITH) programs may accept patients with oxygen saturations at or above 90% or baseline if the etiology of the oxygen requirement is known (e.g., chronic obstructive lung disease, pneumonia), the oxygen requirement is stable, and the patient's clinical condition is either expected to improve or care provided in the HITH is consistent with the patient's goals of care.

5:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

6:

The term "severe and persistent mental illness, cognitive impairment" also called "serious and persistent mental illness," refers to a mental illness or cognitive impairment that is characterized by severe symptoms of prolonged duration that require long-term treatment. The illness causes impaired functioning that interferes significantly with primary activities of daily living (ADLs) and results in an inability to maintain independent functioning without treatment, support, and rehabilitation for a prolonged or indefinite period of time. Examples include, but are not limited to, schizophrenia and other psychotic disorders, bipolar disorder, dementia, Alzheimer's disease, and severe cases of major depression.

7:

InterQual® consultants state there should be careful mental health screening for Hospital in the Home programs and that some programs may accept a patient who has depression if there is no history of suicide attempts or ideation and if the home environment is safe.

8:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

9:

Patients in Hospital in the Home (HITH) programs that include the conditions heart failure, chronic obstructive pulmonary disease (COPD) exacerbations, cellulitis, community acquired pneumonia, and stroke showed comparable mortality and readmission rates when compared with patients treated in traditional inpatient hospital settings (Conley et al., JAMA Intern Med 2016, 176: 1693-702).

10:**Instruction:**

This subset is for patients with a cellulitis infection. For patients with skin infections including: abscess, cutaneous, herpes zoster, wound infection, surgical site infection, and diabetic foot ulcers, refer to Infection: Skin subset. For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

Introduction:

Cellulitis is a diffuse spreading skin infection that extends to the dermis and subcutaneous fat. Clinical manifestations can range from mild to severe and include areas of edema, erythema (redness), warmth, and are often accompanied by lymphangitis. Systemic manifestations can be life-threatening and when symptoms of fever, tachycardia, confusion, hypotension, or leukocytosis are present, it may be indicative of a severe infection. The most frequent organisms causing cellulitis are β -hemolytic streptococci and *Staphylococcus aureus* (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Rrapi et al., Med Clin North Am 2021, 105: 723-35).

Evaluation and treatment:

Most patients with cellulitis can be managed in the outpatient setting with oral anti-infectives and wound management. Hospitalization may be required if the cellulitis fails to respond to outpatient anti-infective treatment, the patient is currently using IV substances, has severe obesity (Body mass index (BMI) greater than 40kg/m²) and unable to manage wound care (e.g., unable to offload, unable to reach or dress wound), or has comorbid conditions (e.g., lymphedema, immunocompromised) associated with poor outcomes.

Intravenous anti-infective treatment for cellulitis is based on history of present illness, risk for resistant organisms, and physical assessment. Specialty consultations with dermatology and infectious disease are recommended if the cellulitis infection is not responding or slow to respond to anti-infective treatment (Boettler et al., Am J Clin Dermatol 2022, 23: 153-65; Rrapi et al., Med Clin North Am 2021, 105: 723-35; Ko et al., JAMA Dermatol 2018, 154: 529-36; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

11:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen.

Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections.

Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases.

Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.

Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e.g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78; Lipska et al., Diabetes Care 2013, 36: 3535-42).

12:

Medical implants are types of medical devices which enhance, support, or replace existing impaired or missing body parts or functions. These devices may be permanent or temporary and composed of man-made or biologic materials. Indwelling (vs transcutaneous) implants pose a greater risk for the development of cellulitis complicated by bacteremia (U.S. Food and Drug Administration, Implants and Prosthetics. 2023).

13:

Cellulitis care includes immobilization and elevation of affected limbs, pain control, warm compresses to affected areas, wound care, and in patients with diabetes, blood sugar control. Marking the erythematous borders present on admission is a valuable method of assessing cellulitis progression or improvement.

14:**Introduction:**

Chronic obstructive pulmonary disease (COPD) is a progressive condition characterized by persistent respiratory symptoms and chronic airflow limitation. The airflow limitation, determined by spirometric abnormality, is not fully reversible and is associated with an increased inflammatory response to noxious gases or particles.

When the patient is diagnosed with COPD, the severity of airflow limitation is classified into 4 stages based on the forced expiratory volume (FEV₁) measurement following bronchodilator administration as follows:

- GOLD 1 Mild: FEV₁ ≥ 80% predicted
- GOLD 2 Moderate: FEV₁ 50-79% predicted
- GOLD 3 Severe: FEV₁ 30-49% predicted
- GOLD 4 Very severe: FEV₁ < 30% predicted

Although several studies have shown a strong correlation between the Global Initiative for Chronic Obstructive Lung Disease (GOLD) classification system and an increased risk of COPD exacerbation, the Combined COPD Assessment provides a more accurate assessment of the patient's level of risk than the GOLD classification alone (Karloh et al., Chest 2016, 149: 413-25). The Combined COPD Assessment includes the patient's GOLD Classification (GOLD 1-4), the patient's perception of breathlessness (using the Modified Medical Research Council (mMRC) questionnaire with mMRC Grades 0-4 or the COPD Assessment Test (CAT) with a scale of 0-40), and the patient's comorbidities.

These factors are used to assign the patient with COPD to 1 of 3 patient groups (A, B, or E) (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

Patient Group A

- Low risk, less symptoms
- Typically GOLD 1 or GOLD 2
- 0-1 exacerbation per year and no hospitalization for exacerbation
- CAT score < 10 or mMRC grade 0-1

Patient Group B

- Low risk, more symptoms
- Typically GOLD 1 or GOLD 2
- 0-1 moderate exacerbation per year and no hospitalization for exacerbation
- CAT score ≥ 10 or mMRC grade ≥ 2

Patient Group E

- High risk
- ≥ 2 moderate exacerbations per year or ≥ 1 hospitalization for exacerbation

A COPD exacerbation is defined as an acute change from baseline dyspnea and/or cough and sputum production which worsens over a period of less than 14 days (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024). Most COPD exacerbations can be attributed to either a respiratory infection or exposure to pollutants, but for many exacerbations the cause is never identified. For patients with unexplained exacerbations, heart failure, pneumonia, or pulmonary embolism should be considered as a cause (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; Aleva et al., Chest 2017, 151: 544-54).

15:

Evaluation and Treatment:

A severe chronic obstructive pulmonary disease (COPD) exacerbation may be indicated by the following signs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- Accessory respiratory muscle use
- Cyanosis, new onset or worsening
- Paradoxical motion of the abdomen
- Retraction of the intercostal spaces

Short-term management goals of an exacerbation include relieving symptoms, minimizing the impact of the current exacerbation, and preventing the development of subsequent occurrences. Long-term goals include improving exercise tolerance, decreasing mortality, preventing disease progression, and decreasing the number of exacerbations. Although a chest radiograph is not always helpful in establishing a diagnosis, it can be a useful tool in ruling out other conditions. Other diagnostic testing may include an electrocardiogram, oxygen saturation, arterial or venous blood gas analysis, complete blood count, blood glucose, and electrolyte panel. During an acute exacerbation, measures of pulmonary function (e.g., FEV₁, spirometry) are not recommended.

Oxygen administration may be required to treat hypoxemia in an acute exacerbation. However, in patients with COPD who retain carbon dioxide (CO₂), oxygen administration may worsen CO₂ retention and, in extreme cases, lead to respiratory arrest. Thus, supplemental oxygen should be administered with careful monitoring of the patient's respiratory and mental status. For patients in acute respiratory failure, the first choice of ventilation should be noninvasive mechanical ventilation, unless contraindicated (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024). Oxygen delivered via high-flow nasal cannula (HFNC) has been found to be useful for some patients with COPD. For a small cohort of hospitalized patients, one study found that its use decreased the patient effort required to maintain adequate oxygenation determined by the measurement of respiratory rate, Pco₂, tidal volume, and patient comfort (Braunlich et al., Int J Chron Obstruct Pulmon Dis 2016, 11: 1077-85).

Pharmacological management of an acute exacerbation should include a systemic corticosteroid and a short-acting inhaled beta-agonist, with or without a short-acting anticholinergic. Therapy with intravenous methylxanthines (e.g., aminophylline or theophylline) is no longer recommended due to severe side effects. Although the use of anti-infectives remains controversial, administration is indicated when there is an increase in dyspnea, sputum volume, sputum purulence, or if mechanical ventilation is required. If there is no response to standard COPD treatment, anti-infectives guided by sputum culture and sensitivity are recommended. Additional guideline recommendations include (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis,

Management and Prevention of COPD. 2024):

- Pulmonary rehabilitation program, initiated during the acute hospitalization, if possible, and at least 6-8 weeks in duration
- Disease management program

Many patients with COPD are at high risk for readmission, and careful attention to discharge planning is essential.

16:

A 2012 Cochrane Database Systematic Review comparing Hospital in the Home (HITH) outcomes versus acute inpatient hospital outcomes for patients with chronic obstructive lung disease (COPD) found moderate quality evidence supporting lower readmission and mortality rates for patients treated in the HITH setting (Jeppesen et al., Cochrane Database Syst Rev 2012: Cd003573).

17:

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

18:

In an effort to address inequities in medicine and improve appropriate access to care, multiple medical societies have been reevaluating the use of adjustments made for a patient's race and updating their calculators and reference equations when reporting diagnostic results.

In 2023, the American Thoracic Society issued updated guidance on reporting of pulmonary function tests (PFTs), advocating for a race-neutral average reference equation to determine the baseline (i.e., predicted normal) for a given patient (Bhakta et al., Am J Respir Crit Care Med 2023, 207: 978-995). The reference equation previously included lower predicted normal values for patients of certain races and ethnicities (in the US, for Black and Asian patients); a practice which has not been found to have biological validity and may lead to misclassification of diagnoses and disease severity in racial or ethnic minorities (Vyas et al., N Engl J Med 2020, 383: 874-882). Since not all spirometer manufacturers and PFT laboratories have switched over to the new race-neutral average reference equation, clinical judgement and consideration of patient-specific factors should be used when interpreting PFT results and deciding whether patients with borderline reported PFT values in the presence of concerning symptoms or signs should have further evaluation or treatment.

19:

The intent of this criteria point is to provide initial treatment with at least 2 doses of inhaled beta-agonist with or without the addition of ipratropium prior to the decision to admit (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024). A licensed medical professional may include an emergency medical technician, nurse, respiratory therapist, or a physician. The frequency of administration will vary based upon clinical response, severity of hypoxemia, and related tachycardia, and is up to the discretion of the treating provider.

20:

According to the available current evidence which is supported by InterQual® peer reviewers, the Acute level of care is appropriate when the patient presents with the following markers of dyspnea severity:

- Unable to take PO
- Hunched over position

21:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

22:

In the setting of dyspnea, patients prefer sitting because they are unable to breathe effectively in the recumbent position.

23:

BAP-65 is a validated tool used to estimate the severity of an acute chronic obstructive pulmonary disease (COPD) exacerbation and the patient's risk for mechanical ventilation and mortality. This risk is increased by elevated blood urea nitrogen, tachycardia, and mental status change (Tabet et al., J Clin Respir Dis Care 2016, 2:).

24:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

25:

Long-term oxygen therapy is defined as requiring oxygen for greater than 15 hours per day. The following are indications for long-term oxygen therapy (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- PaO₂ 55 mmHg (7.3 kPa) or less
- SaO₂ less than 88%(0.88)
- PaO₂ 55-60 mmHg (7.3-8.0 kPa) with pulmonary hypertension, congestive heart failure with peripheral edema, or hematocrit greater than 55%(0.55)

26:

Cor pulmonale is defined as hypertrophy and dilation of the right ventricle resulting from primary pulmonary disease.

27:

Cor pulmonale places patients at higher risk for hospital readmission and death (Echevarria et al., Thorax 2017, 72: 686-93).

28:

New York Heart Association (NYHA) classification for heart failure is defined as follows (Yap et al., Clinical cardiology 2015, 38: 621-8; New York Heart Association, Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis. 1964):

- Class I - No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.
- Class II - Slight limitation of physical activity. Comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.
- Class III - Marked limitation of physical activity. Comfortable at rest. Less than ordinary physical activity causes fatigue, palpitation, dyspnea, or anginal pain.
- Class IV - Inability to carry on any physical activity without discomfort. Symptoms of cardiac insufficiency or of anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.

29:

Patients with chronic obstructive pulmonary disease (COPD) and congestive heart failure (CHF) as a comorbid

condition have greater morbidity and mortality when hospitalized (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

30:

Patients with a history of heart failure, Class III or IV, may be accepted on a case-by-case basis into Hospital in the Home (HITH) programs. InterQual® external peer reviewers generally recommend a hospital setting versus a HITH program for patients with a new diagnosis of heart failure. Patients with new onset heart failure require extensive hospital-based testing.

31:

Radiology imaging should be performed on all patients hospitalized for the management of pneumonia to confirm diagnosis (given the inaccuracy of clinical signs and symptoms alone), to document the extent and severity of pulmonary infiltrates, and rule out complications that may warrant more intensive treatment (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Jokerst et al., J Am Coll Radiol 2018, 15: S240-s51). However, the diagnosis of pneumonia is not solely based on imaging findings due to the high inter-observer variability of chest x-ray readings and the inability to distinguish viral from bacterial pneumonia in some cases. The diagnosis of pneumonia is a combination of clinical symptoms of lower respiratory tract infection (e.g., cough, fever, sputum production, dyspnea, and/or pleuritic chest pain) and the presence of a new infiltrate on imaging that cannot be explained by any other cause (Aliberti et al., Lancet 2021, 398: 906-19; Seo et al., Respiration 2019, 97: 508-17).

The radiologist reading the chest x-ray may not have access to the patient's clinical findings and may not specify a diagnosis of pneumonia when reporting imaging results. The chest x-ray findings are typically described in the report and may include a statement instructing the reader to correlate the findings to the clinical presentation.

32:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

33:

The Global Initiative for Obstructive Lung Disease (GOLD) guideline recommends anti-infective therapy for patients with moderate to severe exacerbations when one or more of the following clinical signs of a bacterial infection are present (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- Increased dyspnea, sputum volume, and sputum purulence
- Increased sputum purulence and either increased dyspnea or sputum volume
- Invasive or non-invasive mechanical ventilation required

Patients with a chronic obstructive pulmonary disease (COPD) exacerbation who receive anti-infectives for a duration of 5-7 days have a shortened recovery time, as well as decreased risk of relapse, treatment failure, and hospital duration.

34:

Instruction: These criteria apply to the administration of a bronchodilator via a nebulizer, an inhaler, or an intravenous route. According to the available evidence, inhalers produce outcomes that are at least equivalent to nebulizers. Further research is needed to determine the optimal inhalation device according to individual patient characteristics and disease state. Education on correct usage is vitally important to ensure effective medication delivery to the lungs (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024; van Geffen et al., Cochrane Database Syst Rev 2016: CD011826).

35:

Evidence in the medical literature to support this criterion or the effectiveness of this intervention or service is mixed or unclear. The criteria point reflects current best evidence and practice. It is the product of a peer review process involving multiple clinicians with diverse expertise in varied practice and geographic settings.

36:

The administration of systemic corticosteroids in patients with a chronic obstructive pulmonary disease (COPD) exacerbation improves lung function, decreases hypoxemia, shortens recovery time, reduces hospital length of stay, and prevents early reoccurrence. Evidence demonstrates daily administration of 40 milligrams of prednisone in patients who have been hospitalized for more than 5 days is not inferior to a 14-day course in terms of early relapse. This shortened regimen is supported by current guidelines. Studies have shown low-dose oral steroids demonstrate equivalent outcomes to intravenous therapy in the treatment of COPD. Nebulized budesonide may be an alternative treatment for exacerbations with similar benefits to intravenous methylprednisolone in select patients, according to the GOLD guidelines (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024).

37:

Introduction:

Heart failure (HF) is a clinical syndrome which occurs when the myocardium becomes unable to pump blood effectively and causes passive congestion of the pulmonary or venous vasculature. HF may start acutely or evolve into a chronic condition. Acute HF is a life-threatening condition and requires urgent treatment.

Major risk factors for developing HF include myocardial infarction (MI), coronary artery disease (CAD), hypertension, valvular heart disease, congenital heart defects, arrhythmias, viral infections, obesity, and exposure to cardiac toxins (e.g., alcohol, tobacco, cocaine). Additionally, certain medications can cause myocardial toxicity or exacerbate underlying myocardial dysfunction and precipitate acute HF (e.g., chemotherapy, nonsteroidal anti-inflammatory drugs (NSAIDs), antidepressants, antihypertensive medications). Acute exacerbation of chronic HF is often precipitated by non-adherence to medications or dietary restrictions (Heidenreich et al., Circulation 2022, 145: e895-e1032; Page et al., Circulation 2016).

The cardinal manifestations of HF are dyspnea upon exertion or at rest, fatigue, weakness, lower extremity edema, rapid or irregular heartbeat, decreased exercise tolerance, persistent cough or wheezing with production of pink, frothy sputum, nocturia, sudden weight gain, anorexia, and chest pain.

The New York Heart Association (NYHA) developed the following classification system:

- **Class I:** No symptoms with ordinary physical activity
- **Class II:** Mild symptoms and slight limitations with ordinary physical activity
- **Class III:** Marked symptoms and limitations even with light physical activity
- **Class IV:** Symptoms at rest with severe limitations

The American College of Cardiology/American Heart Association (ACC/AHA) developed a system to identify structural disease and risk factors that contribute to morbidity and mortality:

- **Stage A:** High risk for HF but without structural heart disease or symptoms
- **Stage B:** Structural heart disease, increased filling pressure, or otherwise unexplained elevated biomarkers (troponin or natriuretic peptide biomarkers) without symptoms of HF
- **Stage C:** Structural heart disease with prior or current symptoms
- **Stage D:** Refractory HF requiring specialized interventions

Patients with established diagnosis of HF may also be classified based on their left ventricular ejection fraction (LVEF), also referred as ejection fraction (EF) (Heidenreich et al., Circulation 2022, 145: e895-e1032):

- **HF with reduced EF (HFrEF):** LVEF $\leq$ 40%
- **HF with improved EF (HFimpEF):** LVEF was $\leq$ 40% previously but follow up measurement now shows LVEF $>$ 40%
- **HF with mildly reduced EF (HFmrEF):** LVEF 41-49%
- **HF with preserved EF (HFpEF):** LVEF $\geq$ 50%

Patients with advanced heart failure (HF) may have a history of the following (Heidenreich et al., Circulation 2022, 145: e895-e1032; Yancy et al., J Card Fail 2017, 23: 628-51):

- Repeated hospitalizations or emergency department (ED) visits for HF in the past year
- Progressive deterioration in renal function with a rise in blood urea nitrogen (BUN) and creatinine
- Weight loss without other cause (e.g., cardiac cachexia)
- Intolerance to renin-angiotensin-aldosterone system inhibitors due to hypotension or worsening renal function
- Intolerance to beta blockers due to worsening HF or hypotension
- Frequent systolic blood pressures less than 90 mm Hg

- Continued NYHA functional class III or IV symptoms despite HF therapy or requiring intravenous inotropes
- Inability to walk one block on level ground due to dyspnea or fatigue
- Recent need to escalate diuretics to maintain volume status, often reaching daily furosemide equivalent dose greater than 160 mg per day or use of supplemental metolazone therapy
- Progressive decline in serum sodium, usually to less than 134 mEq/L
- Frequent shocks from implantable cardioverter defibrillator (ICD)

Evaluation and Treatment:

The term guideline-directed medical therapy (GDMT) has been introduced to include evaluation and treatments that may be involved in the care of patients with congestive HF. This may include clinical and laboratory assessment, diagnostic testing, procedures, and medications. There are 4 classes of medications recommended in GDMT:

- Angiotensin converting enzyme inhibitors (ACEi) or angiotensin receptor blockers (ARB) or angiotensin receptor neprilysin inhibitors (ARNi)
- Beta blockers
- Mineralocorticoid receptor antagonists (MRA)
- Sodium-glucose cotransporter 2 inhibitors (SGLT2i)

Medication used for each patient may differ based on severity of disease and relative contraindications. SGLT2i are a relatively new class of medication and have been found to be associated with reduced risk of hospitalization and cardiovascular mortality, whether or not there is pre-existing type 2 diabetes. Other medications that may also be utilized include diuretics, isosorbide and hydralazine, ivabradine, and digoxin (Heidenreich et al., Circulation 2022, 145: e895-e1032).

The initial evaluation of suspected exacerbation of HF seek to determine the severity of cardiopulmonary instability by assessing the degree of hypoxemia and the hemodynamic status. Standard diagnostic exams and testing may include:

- Physical examination: vital signs, mental status, weight gain, jugular venous distention, breath sounds, heart sounds, peripheral edema
- Laboratory studies: natriuretic peptide biomarker, troponin, blood urea nitrogen (BUN), creatinine, electrolytes, glucose, complete blood count
- Other studies: chest X-Ray (CXR), electrocardiogram (ECG), echocardiogram

B-type natriuretic peptide (BNP) and N-terminal pro-BNP (NT-proBNP) are serum markers released by the cardiomyocytes in response to ventricular stretch and strain. BNP levels may be elevated with acute HF decompensation. Other conditions that can cause elevated natriuretic peptide levels include acute coronary syndrome (ACS), hypertension with left ventricular hypertrophy (LVH), valvular heart disease, atrial fibrillation, pulmonary embolism, pulmonary hypertension, sepsis, chronic obstructive pulmonary disease (COPD), renal dysfunction, or hyperthyroidism. Factors such as age, sex, weight, and renal function can influence BNP levels. BNP values should be used in conjunction with symptoms, physical examination, and radiological findings to aid in the diagnosis of HF.

Treatment of acute congestive HF is focused on hemodynamic stabilization, support of oxygenation or ventilation, and decongestion. For patients requiring hospitalization for acute decompensation, the following medications may be used: intravenous diuretics and vasodilators, GDMT medications, inotropes, and nitrates. Oxygen therapy should be considered for patients with an O₂ saturation less than 90%. Depending on the underlying etiology, other measures to treat refractory HF may include coronary revascularization (e.g., coronary artery bypass graft (CABG), percutaneous coronary intervention (PCI)), heart valve repair or replacement, implantable cardioverter-defibrillator (ICD) or biventricular pacemaker, cardiac resynchronization therapy (CRT), ventricular assistive devices (VAD), and heart transplant (Heidenreich et al., Circulation 2022, 145: e895-e1032; McDonagh et al., Eur Heart J 2021, 42: 3599-726; Yancy et al., J Card Fail 2017, 23: 628-51; Mebazaa et al., Eur J Heart Fail 2015, 17: 544-58). *Many HF patients are at high risk for readmission, and careful attention to discharge planning is essential.*

38:

Treatment of congestive heart failure is focused on hemodynamic stabilization, support of oxygenation or ventilation, and decongestion, which should lead to symptom relief. Medications, including diuretics, and oxygen administration are the cornerstone of initial management. For the majority of patients, a minimum of 2 hours are required to evaluate response to a dose of diuretics and allow for necessary testing.

39:

Physical examination findings such as new onset lower extremity edema, jugular venous distention, hepatomegaly, and sudden weight gain can be characteristics of decompensation in patients with heart failure (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011; Peacock et al., Ann Emerg Med 2006, 47: 22-33).

40:

In patients presenting with acute heart failure (HF) and evidence of fluid overload (by symptoms, physical examination, or radiologic evaluation), inadequate response to the initial intravenous (IV) dose of a loop diuretic indicates a need for the intensification of therapy in order to achieve decongestion and improvement of symptoms. The onset of action of IV loop diuretic occurs within minutes and urine output usually increases measurably within 2 hours. A spot urine sodium less than 50 mmol/L or urine output of less than 150ml per hour at 2 hours (or less than 100-150 ml per hour during first 6 hours) after the administration of a diuretic is considered an inadequate response to treatment (McDonagh et al., Eur Heart J 2021, 42: 3599-726; Felker et al., Journal of the American College of Cardiology 2020, 75: 1178-95; Mullens et al., Eur J Heart Fail 2019, 21: 137-55). A systematic review supported measuring spot urine sodium to determine which patients exhibit inadequate diuretic response (Tersalvi et al., Eur Heart J Acute Cardiovasc Care 2021, 10: 216-23). A post-hoc analysis of the ROSE Acute Heart Failure randomized trial found that a spot urine sodium of less than or equal to 60 mmol/L in patients with acute HF treated with diuretics predicted lower weight loss with diuresis and a prolonged hospital length of stay (Cunningham et al., Clin Cardiol 2020, 43: 43-9).

41:

Instruction: Diuretic here refers to the IV dose given in the emergency room or urgent care prior to decision to admit.

42:

Patients with an increased blood urea nitrogen (BUN), low systolic blood pressure, or elevated creatinine need close monitoring and treatment as these are known markers for poor outcomes in patients hospitalized with decompensated heart failure (Singer Fisher et al., Emerg Med Pract 2017, 19: S1-S2; Elkayam et al., Am Heart J 2007, 153: 98-104; Peacock et al., Ann Emerg Med 2006, 47: 22-33).

43:

When the baseline serum creatinine level is unknown, the Modification of Diet in Renal Disease (MDRD) equation can be used to estimate an expected creatinine value (assuming a normal baseline value for glomerular filtration rate (GFR) of 75ml/min per 1.73 meters squared). The following values demonstrate one method to estimate baseline serum creatinine (eSCr) (with the omission of the race variable) (Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138; Bellomo et al., Crit Care 2004, 8: R204-12):

Male sex assigned at birth:

- Age 20-24 - eSCr 1.3 mg/dL (115 µmol/l)
- Age 25-39 - eSCr 1.2 mg/dL (106 µmol/l)
- Age 40-65 - eSCr 1.1 mg/dL (97 µmol/l)
- Age > 65 - eSCr 1.0 mg/dL (88 µmol/l)

Female sex assigned at birth:

- Age 20-29 - eSCr 1.0 mg/dL (88 µmol/l)
- Age 30-54 - eSCr 0.9 mg/dL (80 µmol/l)
- Age > 54 - eSCr 0.8 mg/dL (71 µmol/l)

Both the MDRD and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations provide a calculation of estimated GFR using serum creatinine, sex assigned at birth, age, and race. Recognition of substantial diversity within population groups has led to multiple organizations no longer including race in the estimated GFR calculations. This approach was investigated by a joint task force of the National Kidney Foundation and the American Society of Nephrology, the result of which was a recommendation to implement the new CKD-EPI creatinine equation which does not use the race variable (Delgado et al., Am J Kidney Dis 2022, 79: 268-88 e1; Inker et al., N Engl J Med 2021, 385: 1737-49).

44:

Patients with acute kidney injury (AKI) and decompensated heart failure (HF) are at risk for poorer outcomes, recurrent admissions, and may quickly advance to chronic kidney disease. Achieving successful diuresis may be challenging and patients with concurrent acute HF and AKI may need advanced therapies (e.g., inotropes). The focus is to achieve optimum decongestion while preserving kidney function (Hollenberg et al., J Am Coll Cardiol 2019, 74: 1966-2011; Bielecka-Dabrowa et al., Curr Heart Fail Rep 2018, 15: 224-38; Takaya et al., Circ J 2015, 79: 1520-5).

45:

Patients hospitalized with acute heart failure (HF) and chronic kidney disease (CKD) are less responsive to loop diuretics and efficacy of established treatments may be reduced because of renal insufficiency. Patients with advanced CKD and acute HF require complicated management to achieve optimal fluid status (Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rangaswami et al., Circulation 2019, 139: e840-e78). A study of over 52,000 patients admitted with acute decompensation of chronic systolic or diastolic heart failure concluded that a serum creatinine level of 2.75 mg/dl or greater on admission predicted a high risk for in-hospital mortality (Fonarow et al., JAMA 2005, 293: 572-80).

46:

Multiple studies have identified an elevated level of blood urea nitrogen (BUN) on admission as a marker of poor prognosis for patients presenting with acute decompensated heart failure (HF) (Peterson et al., Circ Cardiovasc Qual Outcomes 2010, 3: 25-32; Rohde et al., J Card Fail 2006, 12: 587-93; Fonarow et al., JAMA 2005, 293: 572-80). A study of over 52,000 patients admitted with acute decompensation of HF concluded that a BUN level of 43 mg/dl or greater on admission was the single best predictor of in-hospital mortality (Fonarow et al., JAMA 2005, 293: 572-80).

47:

Acute or admission hyponatremia, especially sodium < 130 mEq/L (130 mmol/L) in patients with acute heart failure is associated with increased diuretic needs, decreased renal function, prolonged hospitalization, and increased risk of mortality (Ng et al., Cardiology 2014, 128: 333-42).

48:

The American College of Cardiology and American Heart Association (ACC/AHA) recommend the following labs be monitored during the use of intravenous (IV) diuretics or active titration of heart failure (HF) medications (Heidenreich et al., Circulation 2022, 145: e895-e1032):

- Daily serum electrolytes
- Blood urea nitrogen (BUN)
- Creatinine

49:

Intravenous (IV) loop diuretics (e.g., furosemide, torsemide, bumetanide, ethacrynic acid) are administered to relieve the symptoms of dyspnea and congestion without excessively reducing intravascular volume. Due to their relatively short half-life, diuretic effectiveness can be enhanced by continuous administration or multiple boluses daily. Careful monitoring of daily weights, orthostatic vital signs, intake and output, electrolytes, and renal function are key components of diuretic therapy. If a patient does not initially respond to IV diuretics, other options may be considered, including increasing the dose to ensure adequate drug levels reach the kidneys and adding a second diuretic, typically a thiazide, to the loop diuretic in order to improve diuretic responsiveness (Heidenreich et al., Circulation 2022, 145: e895-e1032; Maddox et al., J Am Coll Cardiol 2021, 77: 772-810; Rosendorff et al., Circulation 2015).

50:**Instruction:**

This subset is for patients with bacterial, fungal, or parasitic community- or healthcare-acquired pneumonia. For patients with known or highly suspected COVID-19, refer to the Infection: COVID-19 subset. For patients with actual or suspected viral respiratory infection, refer to the Infection: General subset. For patients who begin to exhibit signs and symptoms consistent with sepsis and need additional interventions which are not provided in the current subset, please refer to the Infection: Sepsis subset.

Introduction:

Pneumonia is defined as an acute infection of the lung parenchyma. Community-acquired pneumonia (CAP) refers to pneumonia that is acquired outside the hospital setting, whereas healthcare-acquired pneumonia (HAP) refers to pneumonia that occurs 48 hours or more after hospital admission. Clinical manifestations of CAP include cough, increased sputum production, dyspnea, fever, and/or pleuritic chest pain. The following physical exam findings have been reported to be independent predictors of CAP in adults: temperature of 37.8°C (100°F) or higher, crackles on auscultation, oxygen saturations less than 95%(0.95) and pulse greater than or equal to 100 beats/min. Older adults (65 years or older) and immunocompromised patients may exhibit more subtle, non-specific symptoms such as lethargy, change in mental status, dehydration, loss of appetite, and/or worsening of comorbidities, which may occur in the absence of fever. While the causal pathogen is often not identified, the most common bacterial pathogen isolated in CAP is streptococcus pneumoniae. Other common CAP pathogens include Haemophilus influenzae, Moraxella catarrhalis, and Staphylococcus aureus, and atypical organisms such as Legionella, Mycoplasma, and Chlamydophila pneumoniae (Aliberti et al., Lancet 2021, 398: 906-19). The microorganisms responsible for HAP differ based on geographic location; however, gram-negative bacilli such as Pseudomonas aeruginosa (P. aeruginosa) and Enterobacteriaceae are common causal pathogens in HAP (Luyt et al., Curr Opin Crit Care 2018, 24: 332-8).

Evaluation and Treatment:

The diagnosis of pneumonia is established based on the presenting symptoms, physical findings, and is confirmed by the presence of a new pulmonary infiltrate on imaging that is not due to another cause (National Institute for Health and Clinical Excellence (NICE): Pneumonia in adults: diagnosis and management. Clinical Guideline 191. 2019). Although a chest computerized tomography (CT) scan and lung ultrasonography can be used to diagnose pneumonia, a posteroanterior and lateral chest x-rays are the most used imaging techniques. Findings that may be seen on chest x-ray include focal non-segmental or lobar pneumonia, multifocal bronchopneumonia or lobular pneumonia, or focal or diffuse interstitial pneumonia. A chest CT may be needed to confirm pneumonia when there is high suspicion based on clinical findings if the chest x-ray is negative (e.g., in patients who are immunocompromised) (Aliberti et al., Lancet 2021, 398: 906-19).

Pretreatment diagnostic testing, such as gram stain and cultures of lower respiratory secretions and blood cultures, are not indicated except for the following patients (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67):

- Severe disease, especially if intubated
- Empirically treated for P. aeruginosa or methicillin-resistant Staph. aureus (MRSA)
- History of MRSA or P. aeruginosa
- Previously hospitalized and received parenteral antibiotics in the last 90 days (whether during hospitalization or not)

The use of procalcitonin to determine the need for initial antibacterial therapy is not recommended, as the threshold that differentiates between viral and bacterial pathogens in CAP is unknown (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67). Anti-infectives are the mainstay of treatment for patients with bacterial pneumonia. Empiric antibiotics selection should be made according to the most likely pathogen, local microbiology, patient risk factors for specific pathogens, and disease severity (Aliberti et al., Lancet 2021, 398: 906-19). Healthcare-acquired pneumonia is usually caused by gram-negative bacteria which are unique to each facility and initial treatment regimens are guided by local sensitivity patterns (Luyt et al., Curr Opin Crit Care 2018, 24: 332-8). If *Mycobacterium tuberculosis* is determined to be the cause, a multidrug anti-tuberculosis regimen is recommended (Moran and Waxman, In: Walls et al., ed. Rosen's Emergency Medicine: Concepts and Clinical Practice, 9th edn. 2018:871-80). Airborne isolation precautions are required until three consecutive negative AFB sputum smear or gastric aspirate results are obtained. Once clinically stable, discharge may be arranged (even when the smear remains positive) if the patient has begun a multidrug anti-tuberculosis regimen, does not pose a public health risk, and has directly observed therapy (DOT) arranged.

Intravenous antibiotics should be switched to an oral regimen as soon as there is clinical stability and the patient has been afebrile for 24 hours. In a patient with pneumonia, it may take up to 6 weeks for a CXR to show improvement. Follow up CXR should be limited to patients who are not improving or are demonstrating clinical deterioration (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Lim et al., Thorax 2009, 64 Suppl 3: iii1-55).

51:

Imaging studies that can be utilized to aid in the diagnosis of pneumonia include chest x-ray, chest computed tomography (CT), and lung ultrasonography. Although a chest x-ray is the most commonly used imaging study, a chest CT may be required if the chest x-ray is negative but there is a high clinical suspicion for pneumonia (Aliberti et al., Lancet 2021, 398: 906-19). In addition, chest CTs have been found to be superior to chest x-rays in identifying

lesions in the lung parenchyma as well as pleural abnormalities (Seo et al., Respiration 2019, 97: 508-17). The use of lung ultrasound is a newly emerging tool that has shown high accuracy in the diagnosis of pneumonia (Staub et al., J Emerg Med 2019, 56: 53-69).

52:

Multilobar pneumonia places a patient at risk for treatment failure which has been associated with a longer length of stay and an increase in mortality (Peyrani et al., Chest 2020, 157: 34-41; Lim et al., Thorax 2009, 64 Suppl 3: iii1-55). Studies have also identified multilobar pneumonia as an independent risk factor for mortality, and as a result, is a common criterion in community-acquired pneumonia prognostic scoring tools (Smith et al., Ann Emerg Med 2021, 77: e1-e57; Mannu et al., Eur J Intern Med 2013, 24: 857-63).

53:

Community-acquired pneumonia (CAP) can contribute to the worsening condition of patients with existing heart failure (HF) and chronic obstructive pulmonary disease (COPD). Previous studies have failed to show an association between pneumonia in COPD patients and an increase in mortality; however, a recent meta-analysis found that CAP in hospitalized COPD patients was associated with a longer length of stay, increased intensive care unit (ICU) admissions, and higher mortality (Yu et al., Respiration 2021, 100: 64-76). Patients with existing HF have been found to have up to a 4 times higher risk for mortality associated with pneumonia (Shen et al., J Am Coll Cardiol 2021, 77: 1961-73).

54:

The severity of chronic obstructive pulmonary disease (COPD) airflow limitations is classified into 4 grades based on the patient's baseline forced expiratory volume (FEV₁) measurement post-bronchodilator administration as follows (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- GOLD 1 - Mild: FEV₁ ≥ 80%(0.80) predicted
- GOLD 2 - Moderate: FEV₁ 50-79%(0.50-0.79) predicted
- GOLD 3 - Severe: FEV₁ 30-49%(0.30-0.49) predicted
- GOLD 4 - Very severe: FEV₁ < 30%(0.30) predicted

55:

Heart failure (HF) patients with hypertension, diabetes mellitus, obesity, and metabolic syndrome require complex management. Treating comorbidities adequately can improve the outcomes of a HF exacerbation (Bozkurt et al., Circulation 2016, 134: e535-e78).

56:

Expected treatments for patients with bacterial pneumonia include anti-infective and oximetry or blood gas monitoring (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; Cohoon et al., Am J Med 2018, 131: 307-16 e2; Kaysin and Viera, Am Fam Physician 2016, 94: 698-706; Jain et al., N Engl J Med 2015:[e-pub]).

57:

When a patient is admitted through the emergency department (ED), the initial dose of anti-infective is routinely administered while still in the ED. Guidelines support intravenous antibiotic therapy as the initial treatment for most patients requiring inpatient care, especially those with severe disease. There is a role for oral antibiotics in the inpatient setting to treat a subset of patients with less severe disease whose clinical findings still support admission, but the decision to administer oral antibiotics does not drive the need for inpatient treatment (Metlay et al., Am J Respir Crit Care Med 2019, 200: e45-e67; National Institute for Health and Clinical Excellence (NICE): Pneumonia in adults: diagnosis and management. Clinical Guideline 191. 2019; Lee et al., Jama 2016, 315: 593-602).

58:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

59:

Acute Pyelonephritis is an infection of the upper urinary tract, including the renal parenchyma and renal pelvis. Uncomplicated infection is defined by a typical urinary bacterium in an immunocompetent patient with normal urinary tract anatomy and renal function. Characteristics that define complicated pyelonephritis include immunocompromised patients or those with altered urinary tract anatomy, such as patients with strictures, tumors or other obstructions to urine flow. The clinical presentation of pyelonephritis may include fever, chills, flank pain, nausea, vomiting, or costovertebral angle tenderness. Most patients with pyelonephritis can be managed in the outpatient setting. Acute hospital level of care may be necessary for those with complicated pyelonephritis or who have not responded to outpatient therapy (Johnson and Russo, N Engl J Med 2018, 378: 48-59; Takhar and Moran, Infect Dis Clin North Am 2014, 28: 33-48).

Evaluation and Treatment

The diagnosis of pyelonephritis is confirmed with a positive urinalysis and urine culture, along with clinical findings. The use of routine blood cultures is not recommended. Blood cultures should be obtained when the patient has an uncertain diagnosis, is immunocompromised, or has a suspected hematogenous infection (Johnson and Russo, N Engl J Med 2018, 378: 48-59).

Patients who require Acute hospital level of care should receive an intravenous anti-infective. Most patients respond to treatment within 72 hours. If there is no improvement, a change to the anti-infective regimen and imaging are recommended. Renal ultrasound, although useful in determining if the patient has abnormal urinary tract anatomy without a radiation risk, is not as accurate as computed tomography (CT) of the pelvis and abdomen in diagnosing pyelonephritis (American College of Radiology, ACR appropriateness criteria: acute pyelonephritis. 2012. Revised 2018.).

60:

This criteria point is intended to exclude patients who are chronically colonized at baseline, presenting without urinary symptoms. Patients with altered mental status and no obvious urinary symptoms by history should be evaluated for costovertebral angle (CVA) or suprapubic tenderness on examination (Long and Koyfman, Emerg Med Clin North Am 2018, 36: 685-710). Patients not demonstrating one of the listed urinary symptoms may be appropriate for treatment of suspected urinary tract infection, but would require secondary review as this should be considered on a case by case basis. Asymptomatic bacteriuria is a common finding across all populations, with notable variations by age, gender, presence of genitourinary abnormality, or other factors.

61:

Urinary symptoms may include:

- Costovertebral angle (CVA) pain or tenderness
- Dysuria
- Frequency
- Flank pain
- Hematuria, gross
- Incontinence
- Suprapubic pain, pressure, or tenderness
- Urgency

62:

A WBC count greater than $10/\text{mm}^3$ in a urinalysis usually indicates a bacterial infection in the urinary tract (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Centers for Disease Control and Prevention, Urinary Tract Infection Events. 2015).

63:

A positive leukocyte esterase on the urinalysis indicates WBCs in the urine (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Centers for Disease Control and Prevention, Urinary Tract Infection Events. 2015).

64:

Positive nitrites in the urinalysis are a highly sensitive (95-98%) indicator for urinary tract infection (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23).

65:

Pyelonephritis or complicated urinary tract infection in pregnancy may require hospitalization due to the associated dehydration, nausea and vomiting, and the potential for preterm labor. Also, there is the uncommon but serious risk of endotoxic shock and endotoxic damage to alveolar capillary membranes which leads to pulmonary edema and the potential for acute respiratory distress syndrome. Patients who present at 24 weeks gestation or greater are at higher risk for complications related to pyelonephritis or complicated urinary tract infection. Hospitalization is recommended for management of patients who have moderate to severe symptoms (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Wing et al., Am J Obstet Gynecol 2014, 210: 219 e1-6).

66:

Patients with urinary stents or other indwelling devices, such as nephrostomy tubes or permanent foley catheter are at increased risk for complicated urinary tract infection as they have a higher risk of treatment failure and sepsis (Kozyrakis et al., J Endourol 2020, 34: 516-22; Petrosillo et al., Expert Rev Anti Infect Ther 2020, 18: 47-61).

67:

Patients with a urinary tract infection due to a urinary obstruction are at increased risk for sepsis. Examples of urinary tract obstruction include malignancy and renal calculi. Anatomical obstruction may be due to hydronephrosis or urinary fistula (Kozyrakis et al., J Endourol 2020, 34: 516-22; Petrosillo et al., Expert Rev Anti Infect Ther 2020, 18: 47-61).

68:

There is a role for oral antibiotics in the inpatient setting to treat the subset of patients with less severe disease whose clinical findings still support admission, but the decision to administer oral antibiotics does not drive the need for inpatient treatment.

69:

Patients with pyelonephritis, or complicated urinary tract infection, requiring hospitalization should be initiated on IV antibiotics. The following parental anti-infectives are recommended (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Gupta et al., Clin Infect Dis 2011, 52: e103-20. Reaffirmed 2013):

- Fluoroquinolone
- Aminoglycoside with or without ampicillin
- Extended spectrum cephalosporin with or without an aminoglycoside
- Carbapenem

70:

Instruction: To perform a continued stay review for a patient in a hospital in the home program, refer to the Acute level of care in the appropriate condition specific subset for the episode day being reviewed.